

Machine Learning-based Visual Tool for News Reliability Assessment

A Vietnamese NLP/ML web application for reliability screening, visual explanation, history, and feedback collection.

Student

Pham Ngoc Huy Hoan
Trinh Xuan Hoa

Supervisor

Nguyen Ngoc Huyen Tran, M.Sc.

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Vietnamese misinformation and clickbait need screening support.

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Streamlit app integrates inference, explanations, storage, and feedback.

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Demo

Two prepared cases prove reliable vs suspicious behavior and history.

Problem and project boundary

The project screens linguistic reliability signals in Vietnamese news.

PROBLEM

Vietnamese users face unreliable news, clickbait headlines, and posts without clear sources.

A binary fake/real label alone is hard to trust.
Users need risk, signals, and limitations.
Reviewers need history and feedback for later inspection.

BOUNDARY

Decision support for initial screening — not a claim-level truth verification system.

Focus: Vietnamese text reliability patterns.
ML/NLP baselines with measurable evaluation.
Explanation, prediction history, feedback, and report export.

Objectives and contributions

Built around reproducibility, explainability, and feedback.

OBJECTIVES

Build a web-based Vietnamese news reliability screening tool.
Train and compare supervised ML baselines for fake/clickbait detection.
Visualize risk, model signals, suspicious terms, and prediction history.
Collect feedback as a future retraining source.

CONTRIBUTIONS

Reproducible data pipeline: download, clean, split, train, evaluate.
Four-model benchmark with a metric-based best-model decision.
Explainable assessment: ML risk, lexical risk, highlights, token contribution.
Layered architecture: Streamlit, NLP/ML core, Supabase/PostgreSQL.

Dataset and label convention

VFND is normalized into a binary reliability screening task.

PRIMARY SOURCE

VFND Vietnamese Fake News Dataset

Open research dataset for Vietnamese fake-news studies.
Original labels normalized into reliable vs unreliable.
Source link is opened during defense for traceability.

0 = reliable / real news
1 = unreliable / fake / clickbait

350

train samples

75

validation samples

75

test samples

Data preparation evidence

Cleaning is part of the contribution, not a hidden step.

31.32%

duplicate ratio removed

20

minimum length threshold

251

label 0 count

249

label 1 count

CLEANING STEPS

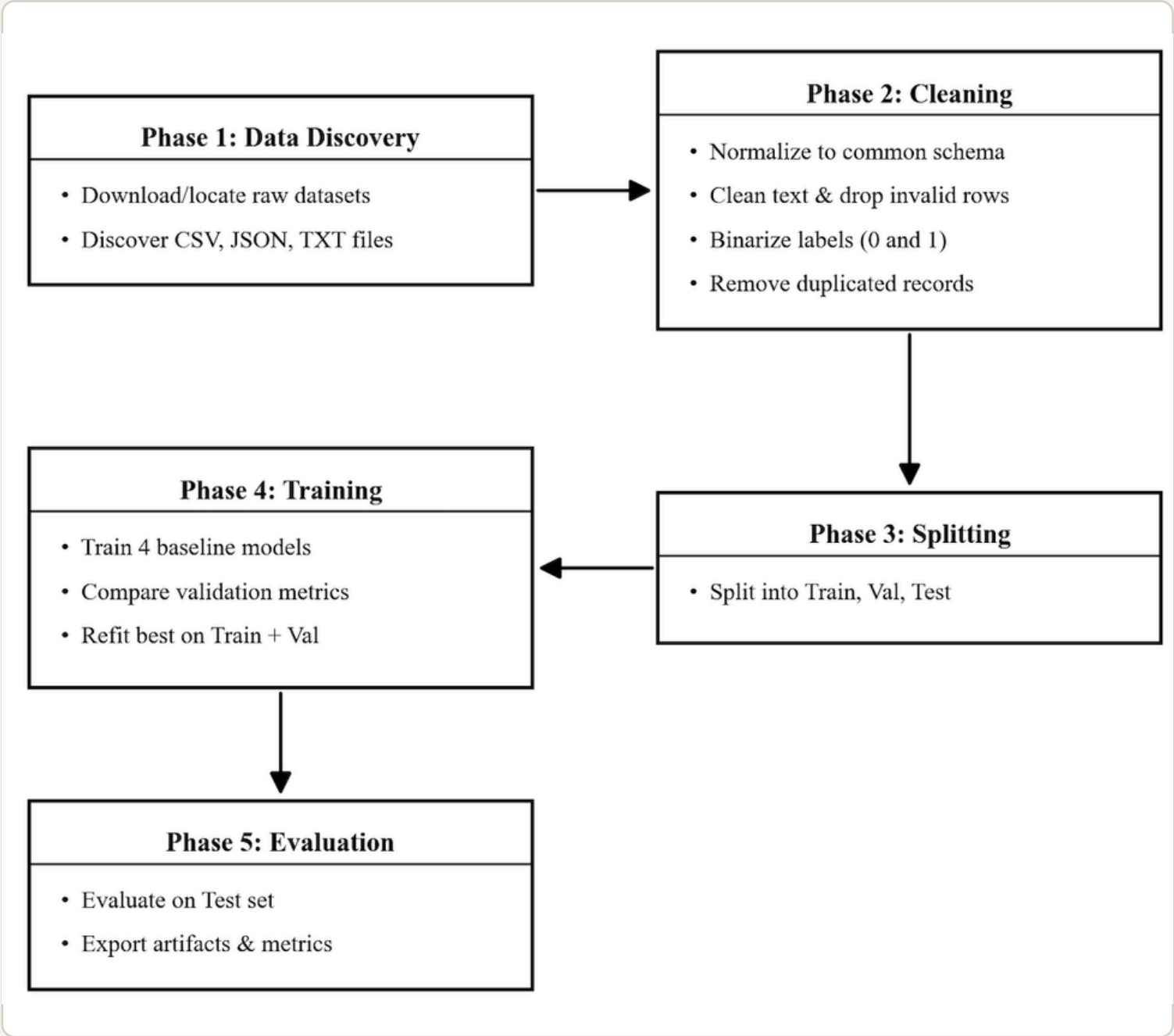
Remove missing labels and unusable text.
Drop duplicates to reduce leakage.
Keep only records passing minimum text length.
Store processed train.csv, val.csv, and test.csv.

WHY IT MATTERS

Clear split prevents training on the final test data.
Balanced labels make F1 macro and accuracy meaningful.
Dataset profile supports academic traceability.

NLP and ML pipeline

From raw dataset to exported model artifact.



- | | | |
|---|-----------------|--|
| 1 | Download | VFND dataset and optional manual source. |
| 2 | Clean | Normalize labels, remove duplicates and short texts. |
| 3 | Split | Train / validation / test. |
| 4 | Train | TF-IDF + four supervised baselines. |
| 5 | Evaluate | F1 macro, accuracy, ROC-AUC, confusion matrix. |
| 6 | Export | best_model.joblib and metadata. |

Methodology choices

Classical baselines are explainable, stable, and defensible.

FEATURE EXTRACTION

TF-IDF

Fast on limited data.
Good for sparse text vectors.
Supports token contribution analysis.
Easier to explain than deep embeddings.

MODEL BENCHMARK

Logistic Regression

Probabilistic linear baseline

Linear SVM

Strong margin classifier for TF-IDF

Random Forest

Tree ensemble comparison

Naive Bayes

Fast classical text baseline

Selection rule: validation first, then final refit and test evaluation.

Model benchmark

The best model is selected by validation F1 macro, not by preference.

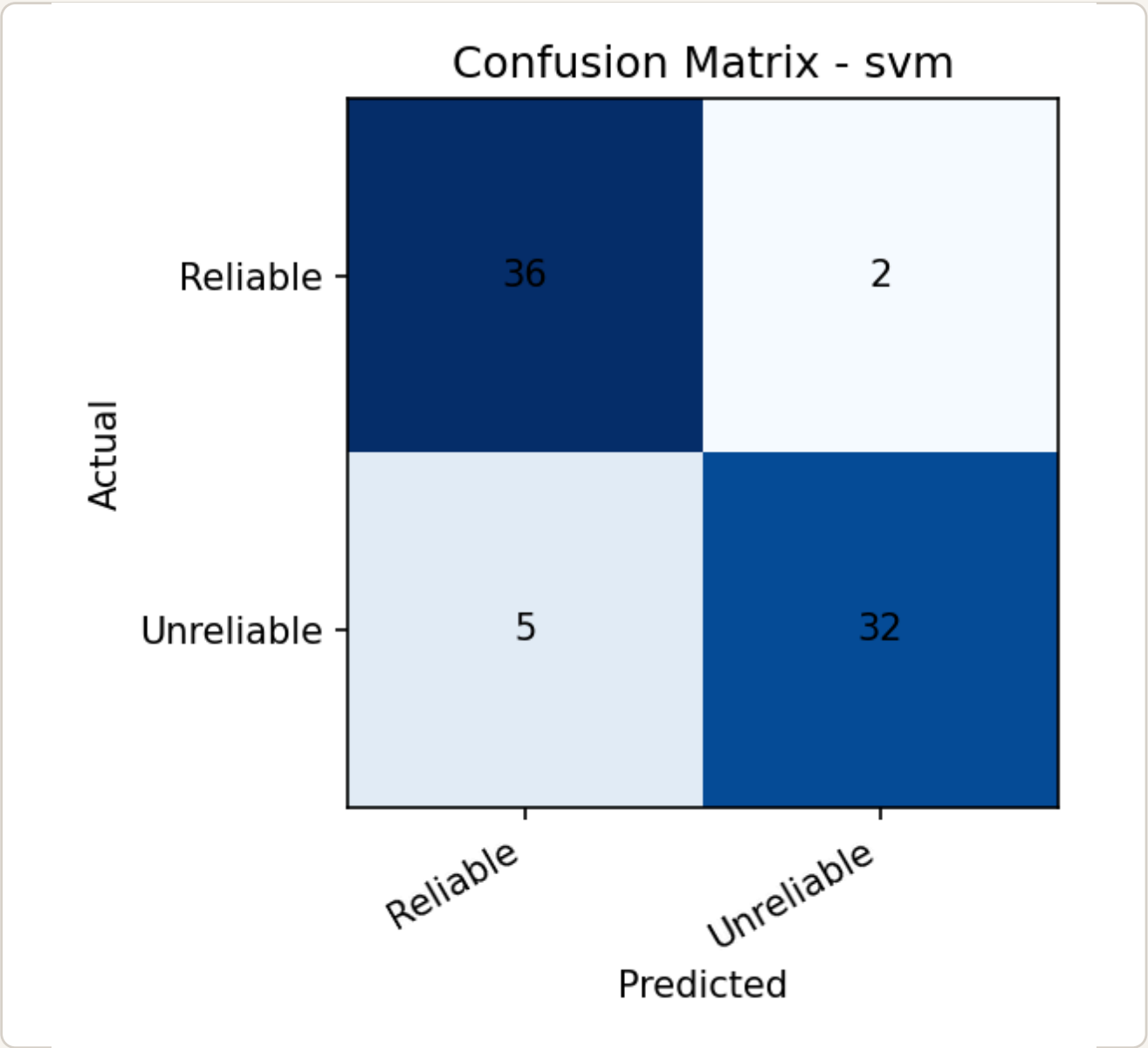
Model	Val Acc	Val F1	Test Acc	Test F1
Logistic Regression	0.9200	0.9196	0.8933	0.8929
Linear SVM	0.9467	0.9466	0.9067	0.9064
Random Forest	0.9200	0.9196	0.8933	0.8929
Naive Bayes	0.9067	0.9061	0.8933	0.8929

Best model evaluation

Confusion matrix supports transparent error discussion.

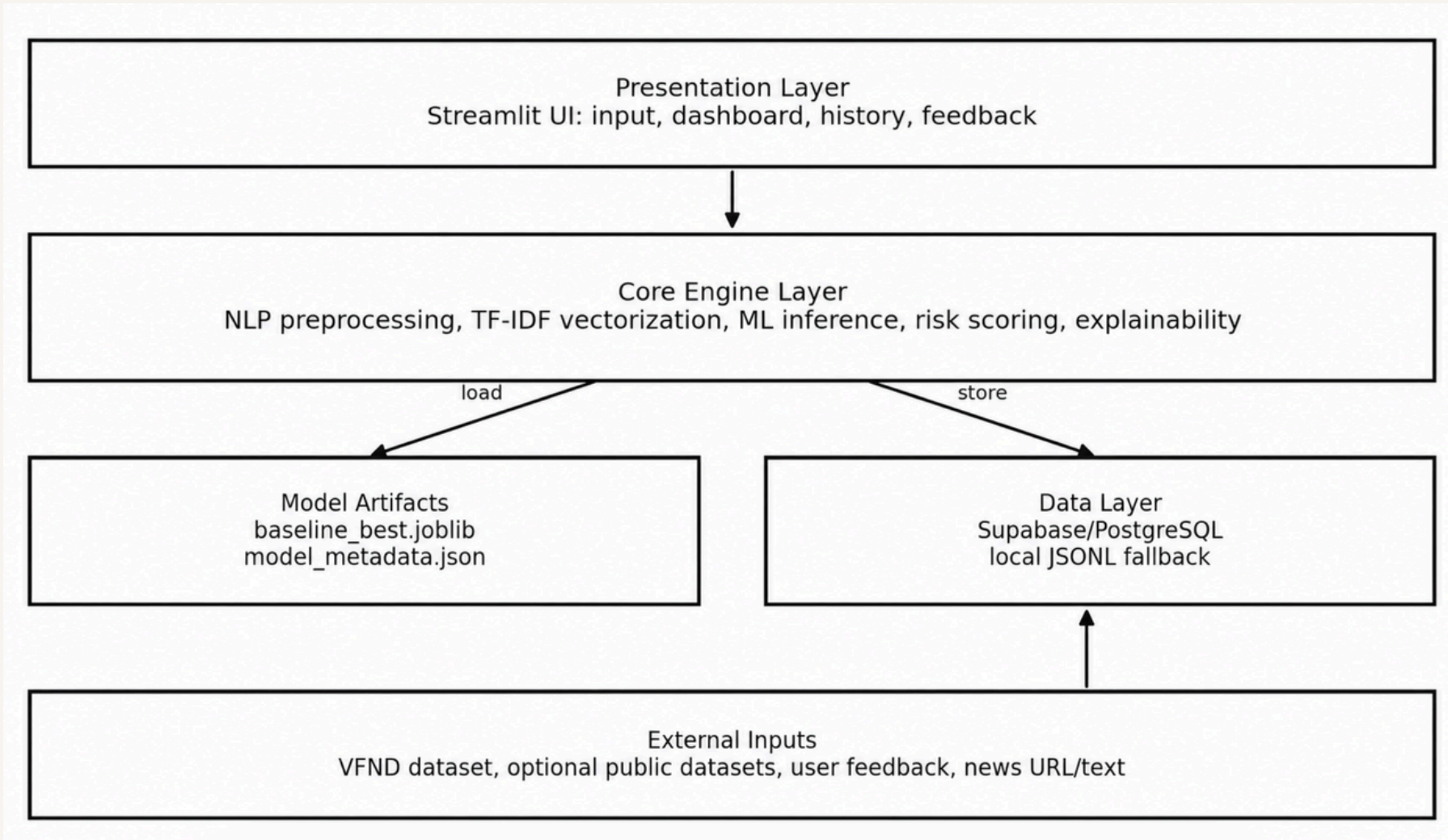
Why Linear SVM works here

TF-IDF creates sparse, high-dimensional vectors.
Linear SVM is strong in this feature space.
Macro F1 handles both labels fairly.



Layered architecture

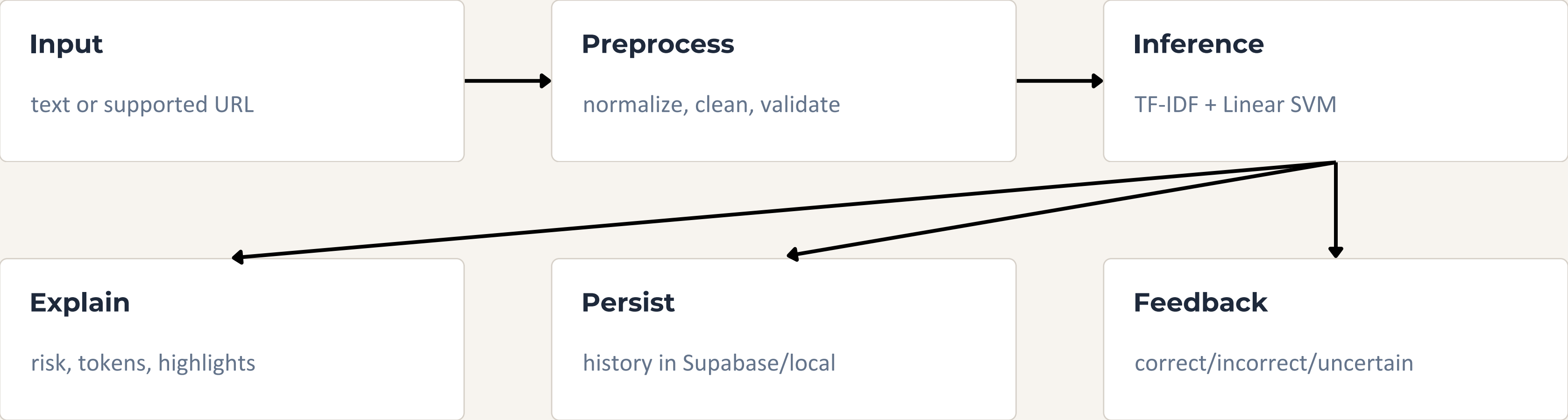
UI, ML core, model artifacts, and storage are separated.



Presentation	Streamlit UI, input, dashboard, history, feedback
Core engine	Preprocessing, TF-IDF, inference, risk scoring, explanation
Data layer	Supabase/PostgreSQL and local fallback storage
Artifacts	best_model.joblib and metadata

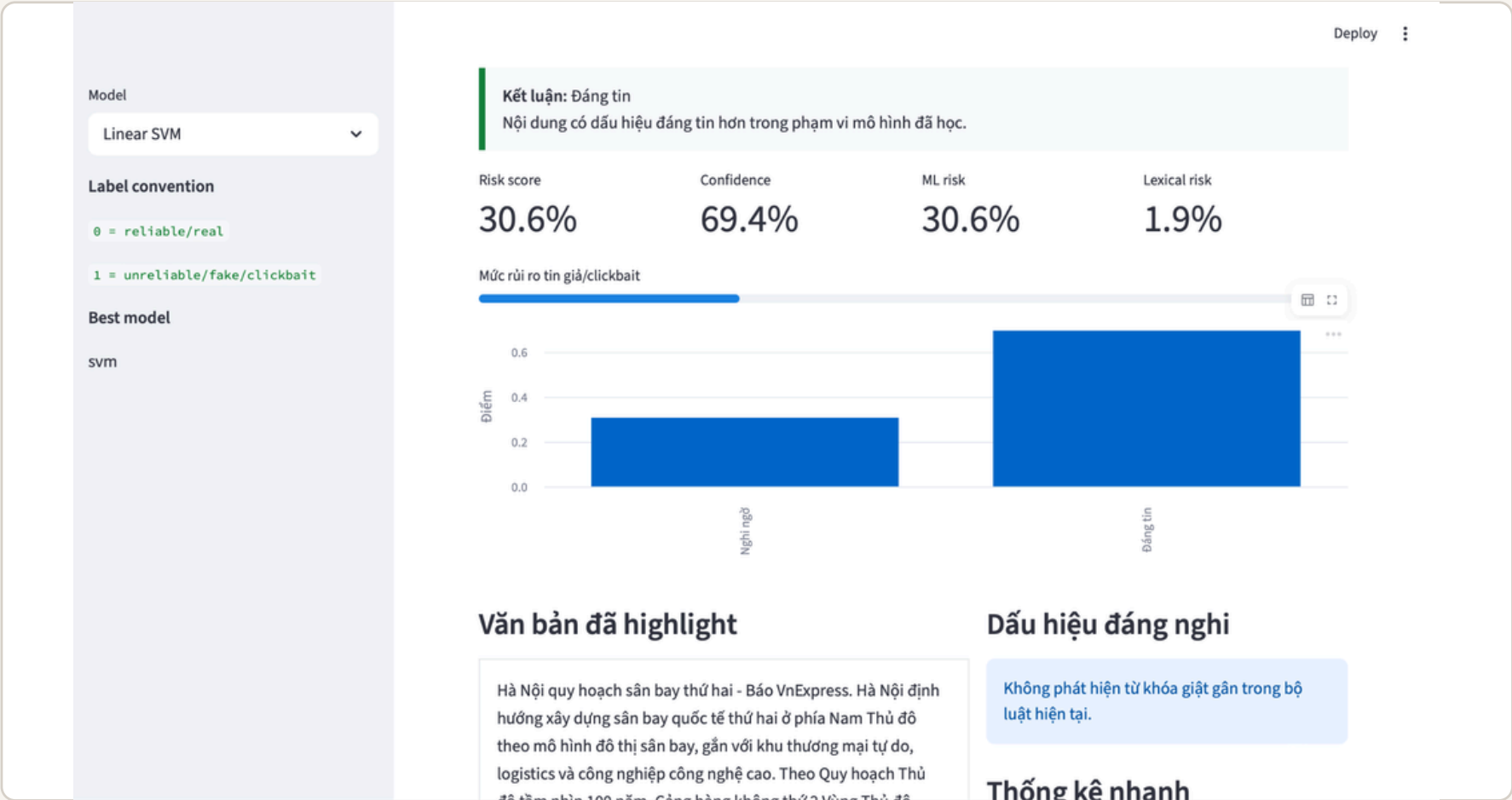
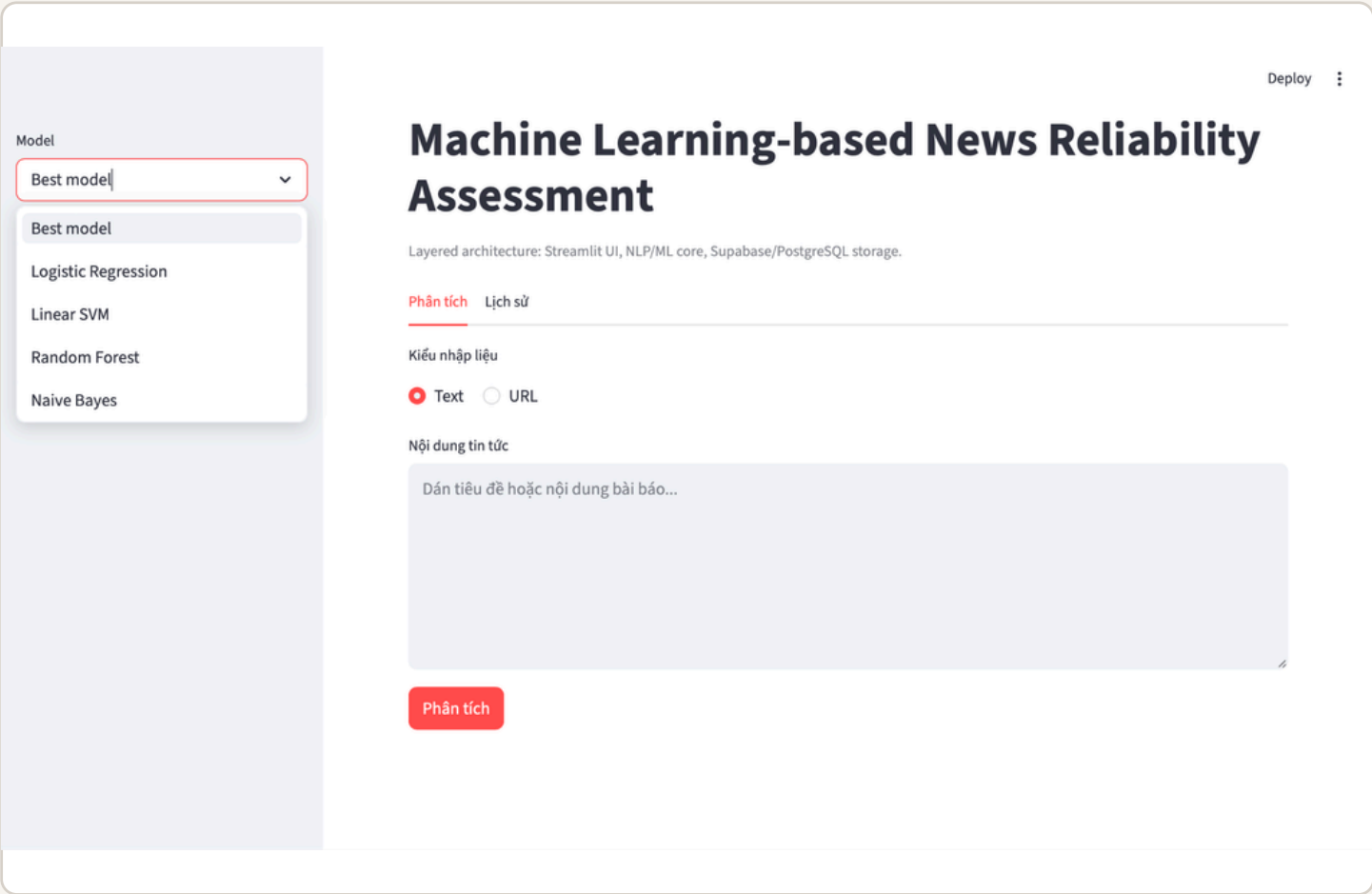
Prediction workflow

The app turns raw input into prediction, explanation, storage, and feedback.



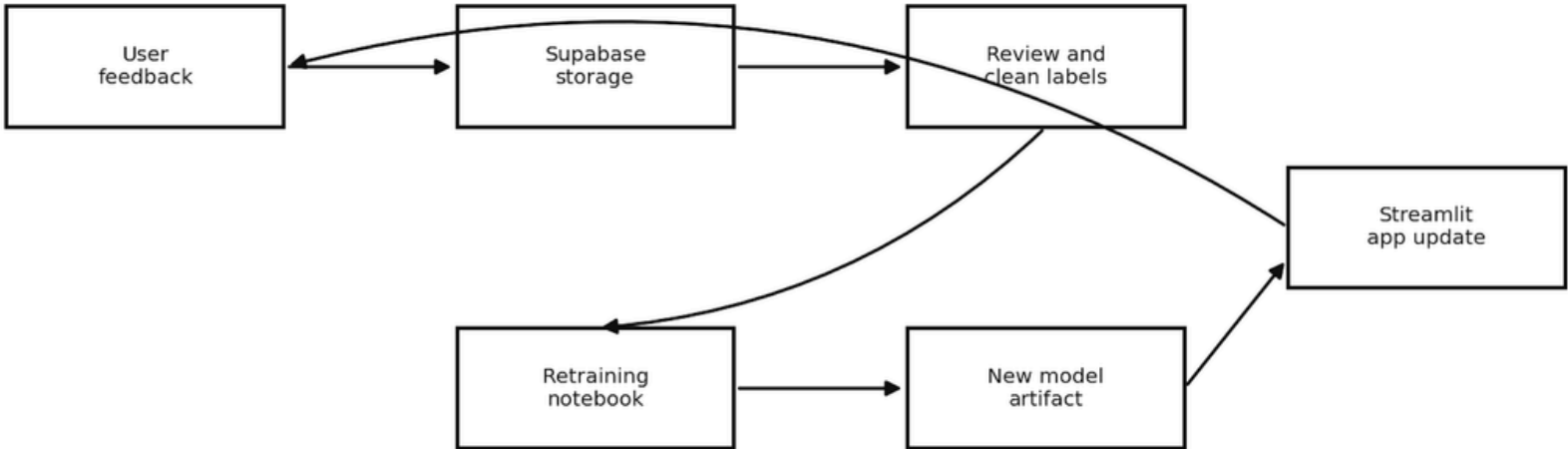
Application result and explainability

The result page is designed for readable review, not just prediction.



Database and feedback loop

Prediction history and user feedback make the workflow reusable.



STORED RECORDS

Predictions: input text, model output, risk score, explanation, created time.
Feedback: user judgment, optional note, linked prediction id.
History: reviewer inspection and future retraining data.

Limitations and future work

Limitations are stated honestly to make the defense stronger.

CURRENT LIMITATIONS

Dataset size is still limited.
The tool screens linguistic reliability signals, not absolute truth.
No claim-level evidence retrieval yet.
Source credibility is not fully modeled.

FUTURE WORK

Expand and continuously update Vietnamese datasets.
Fine-tune PhoBERT or other Vietnamese language models.
Add source credibility and claim evidence retrieval.
Use feedback data for monitored retraining.

Conclusion

A reproducible, explainable, and extensible Vietnamese NLP/ML system.

Working app

Streamlit web application for Vietnamese news reliability screening.

ML pipeline

Dataset cleaning, TF-IDF, four baselines, metrics, exported model.

Explainability

Risk score, lexical signals, highlighted terms, token contribution.

Storage loop

Supabase/PostgreSQL history and feedback collection.

Thank you!